Cyber Security Monitoring System

Report

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Submitted to

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Certificate

This is to certify that Gurveer Singh and Drishti Paras, a student of B.Tech Computer Science, has successfully completed the project titled “Survey Design and Statistical Analysis for Social Data” as a part of academic requirements.

This project has been carried out under my guidance and supervision during the academic year 2026. The work presented in this project report is original and has been completed by the student with sincerity and dedication.

To the best of my knowledge, this project fulfils the requirements prescribed by the institution.

Project Guide Signature:

Name of Guide:

Date:

Institution Name: Chandigarh University

Acknowledgement

I would like to express my sincere gratitude to my teacher for providing me with the opportunity to work on this project titled “Cyber Security Monitoring System.” Their valuable guidance, support, and encouragement helped me successfully complete this project.

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INTRODUCTION

In the modern digital era, cyber security has become one of the most important aspects of information technology. With the rapid growth of the internet, cloud computing, and smart devices, the risk of cyber-attacks has increased significantly. Organizations and individuals are constantly exposed to threats such as hacking, data theft, phishing, and unauthorized access, which can lead to serious consequences like financial loss, data leakage, and system failure.

One of the major vulnerabilities in any system is the presence of open ports. Ports are communication endpoints through which data is transmitted between systems. If these ports are left unsecured or unnecessarily open, attackers can exploit them to gain unauthorized access. Therefore, identifying and analyzing open ports is a critical step in securing any network.

This project presents an AI-Based Cyber Security Monitoring System designed to detect such vulnerabilities in a simple and effective manner. The system uses the Nmap tool to scan a target IP address, identify open ports, evaluate risk levels, and provide alerts to the user. It also includes a user-friendly dashboard built using Streamlit, which allows users to interact with the system and view results in real time along with graphical representations and logs.

Key Highlights:

- Real-time network scanning
- Detection of open ports and services
- Risk level classification (Low, Medium, High)
- Interactive dashboard interface
- Logging and reporting system

The main aim of this project is to provide a practical understanding of cyber security concepts and demonstrate how simple tools can be used to build an effective monitoring system.

OBJECTIVES

The primary objective of this project is to design and develop a cyber security monitoring system that can identify vulnerabilities and provide real-time insights into system security.

Main Objectives:

- To implement network port scanning using Nmap
- To detect open ports and active services on a target system
- To analyse scan results and determine risk levels
- To develop an interactive and user-friendly dashboard
- To store scan results using a logging system
- To visualize data using graphs and charts
- To implement basic malware detection using SHA-256 hashing

This project focuses on the practical implementation of cyber security concepts and aims to provide a simple and effective solution for monitoring system security.

METHODOLOGY

The development of this project follows a systematic and well-structured methodology to ensure accuracy, efficiency, and proper execution of all components. Each stage of the system is carefully designed and implemented to achieve reliable results and smooth functionality.

Step 1: Environment Setup

- Kali Linux was installed using VirtualBox to create a secure and isolated working environment.
- Python environment was configured for development.
- Required tools, libraries, and dependencies were installed and verified.

Step 2: Tool Integration

- The Nmap tool was integrated with Python using the subprocess module.
- Automated execution of scanning commands was implemented to simplify the process.

Step 3: Data Collection

- Network scan results were collected directly from Nmap output.
- Raw data was stored temporarily for further processing and analysis.

Step 4: Data Processing

- The scan output was analysed line by line.
- Relevant information such as open ports was extracted using string filtering techniques.

Step 5: Risk Analysis

- Risk level was calculated based on the number of open ports detected.
- The system classifies risk into three categories:
    - **Low Risk:** No open ports detected
    - **Medium Risk:** 1–3 open ports

- **High Risk:** More than 3 open ports

Step 6: Dashboard Development

- A user-friendly dashboard was developed using Streamlit.
- Input fields and output sections were designed for smooth user interaction.
- Real-time results, graphs, and logs were displayed for better understanding.

Step 7: Testing and Debugging

- The system was tested under different scenarios to ensure reliability.
- Errors were identified and resolved through debugging.
- Performance was optimized for faster and efficient execution.

The above methodology ensures that the system operates effectively, delivers accurate results, and provides a seamless user experience.

TOOLS AND TECHNOLOGIES

This project uses modern tools and technologies for efficient network scanning, data processing, and visualization.

Tools:

Nmap (Network Mapper)

- Used for scanning open ports and services on a target system
- Provides detailed network information

Kali Linux

- Specialized operating system for cyber security tasks
- Includes various pre-installed security tools

VirtualBox

- Used to run Kali Linux in a virtual environment
- Ensures safe testing without affecting the host system

Technologies:

Python

- Main programming language used for development
- Easy to use and supports tool integration

Streamlit

- Used to build the interactive dashboard
- Provides real-time input and output display

Pandas

- Used for data handling and visualization
- Helps in generating graphs

Hashlib

- Used for cryptographic hashing
- Generates SHA-256 hash for file integrity

SYSTEM ARCHITECTURE

The system architecture defines how different components of the system interact and work together to perform cyber security monitoring efficiently. It is designed in a layered structure to ensure smooth data flow, better organization, and easy scalability.

Layers of Architecture:

1. Input Layer

- Allows the user to enter the target IP address
- Provides options to select different scan types
- Acts as the entry point for user interaction

2. Processing Layer

- Core part of the system where computation takes place
- Nmap performs network scanning on the given IP address
- Python processes the scan results and extracts open ports
- Risk analysis is performed
- File hashing (SHA-256) is applied for basic malware analysis

3. Output Layer

- Displays scan results on the dashboard
- Shows risk level, graphs, and logs
- Provides alerts based on detected risks

Benefits of Architecture:

- Simple and easy to understand design
- Efficient data flow between components
- Modular and easy to maintain
- Scalable for future improvements

IMPLEMENTATION

The implementation phase involves the development of multiple modules that work together to build a complete and functional cyber security monitoring system. Each module is designed to perform a specific task, making the system organized and easy to manage.

Modules:

1. Port Scanning Module

- Executes Nmap commands using Python
- Scans the target system for open ports and services
- Collects raw scan results for further processing

2. Data Processing Module

- Analyses the scan output line by line
- Extracts important information such as open ports
- Filters and organizes the required data

3. Risk Analysis Module

- Determines the security level of the system
- Uses conditional logic based on the number of open ports
- Classifies risk into Low, Medium, and High

4. Dashboard Module

- Displays results using an interactive interface
- Allows users to enter input and view output in real time
- Improves user experience with a simple design

5. Logging Module

- Stores scan results in a file
- Maintains history of previous scans
- Helps in tracking and analysis

FEATURES

This system includes several powerful features that make it effective, user friendly, and suitable for real-world applications.

Key Features:

- **Real-Time Port Scanning**
  Scans the target system instantly and detects open ports
- **Risk Level Detection**
  Automatically classifies system security into Low, Medium, or High risk
- **Multi-Scan Modes**
  Allows users to choose between different scanning types
- **Interactive Dashboard**
  Provides a simple and attractive interface for users
- **Graphical Data Representation**
  Displays scan results using graphs for easy understanding
- **Logging System**
  Saves scan results for future reference
- **Downloadable Reports**
  Allows users to download scan results as a report file
- **Malware Hashing (SHA-256)**
  Generates hash values for file integrity and basic analysis
- **Alert System**
  Notifies users when high-risk activity is detected

RESULTS AND DISCUSSION

The developed system was tested under various scenarios to evaluate its performance, accuracy, and reliability. The results obtained were consistent and demonstrated that the system is capable of effectively identifying network vulnerabilities. **Observations:**

- Open ports were detected accurately during scanning
- Risk levels were correctly calculated based on scan results
- The dashboard displayed real-time output without delay
- Logs were successfully generated and stored for each scan

Performance Analysis:

- The scanning process was fast and efficient
- The user interface was smooth and responsive
- The system handled multiple scans without errors
- Logging and data handling were performed efficiently

Advantages:

- Simple and easy-to-understand implementation
- User-friendly and interactive dashboard
- Real-time monitoring and quick results
- Practical application in basic network security analysis

Limitations:

- Risk analysis is based only on the number of open ports
- No integration with real-time malware detection databases
- Limited to basic scanning and analysis features

Overall, the system performed successfully and achieved all the intended objectives. It provides a reliable solution for basic cyber security monitoring and demonstrates practical implementation of key concepts.

CONCLUSION

This project successfully demonstrates the design and development of a cyber security monitoring system using Python and Nmap. It integrates network scanning, data analysis, and visualization into a single platform, making it both effective and easy to use. The system is capable of identifying open ports, analysing potential risks, and presenting the results in a clear and interactive manner.

Key Achievements:

- Implementation of real-time network monitoring
- Development of an automated risk detection system
- Creation of an interactive and user-friendly dashboard
- Successful integration of multiple technologies

In addition to these achievements, the project also highlights how simple tools and techniques can be used to build a practical cyber security solution. The use of visualization and logging improves the overall usability and helps in better understanding of system behaviour. The modular design of the system also makes it easy to maintain, upgrade, and extend with new features.

Furthermore, this project demonstrates the importance of proactive security monitoring and how early detection of vulnerabilities can help prevent potential cyber-attacks. It also reflects the practical application of theoretical concepts learned in cyber security.

The project not only fulfils its objectives but also provides a clear understanding of cyber security fundamentals and their practical applications. It serves as a strong foundation for building more advanced and intelligent security systems in the future.

FUTURE SCOPE

Although the system performs efficiently, there are several opportunities to enhance its capabilities and make it more advanced, intelligent, and industryready. By integrating modern technologies and advanced security features, the system can be transformed into a more powerful and reliable cyber security solution.

Future Enhancements:

- Integration with VirusTotal API for real-time malware detection
- Implementation of AI/ML-based threat prediction systems
- Addition of real-time alert systems such as Email or SMS notifications
- Deployment on cloud platforms for remote access and scalability
- Advanced vulnerability scanning with detailed and automated reporting

In addition to these improvements, the system can also be extended to support multi-user access, role-based authentication, and centralized monitoring for larger networks. Enhancing the user interface and adding more detailed analytics can further improve usability and decision-making.

These enhancements can significantly improve the overall performance and effectiveness of the system. They will not only increase the accuracy of threat detection but also make the system more scalable, automated, and suitable for real-world cyber security environments.